📊 Amazon Sales Insights Report

📌 Problem Statement

The objective of this analysis is to evaluate e-commerce sales data to identify revenue trends, category performance, customer behavior, and the impact of discounts and ratings on business performance. The goal is to generate actionable insights to improve profitability and decision-making.

📊 Dataset Overview

- Total Orders: 50,000
- Total Revenue: $32.87M
- Average Order Value: $657.33
- Time Period: 2022–2023
- Categories: 6
- Regions: 4
- Payment Methods: 5

🔍 Key Insights

📈 Revenue Trends

- Total revenue reached **$32.87M** across 50,000 orders
- Year-over-Year growth is **+0.54%**, showing stable performance
- January is the highest revenue month (~$1.46M), indicating seasonal demand
- February consistently shows the lowest sales (post-holiday drop)
- Revenue is evenly distributed across months with no heavy spikes

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🛍️ Category Performance

- All categories contribute almost equally (~16–17% each), showing a **balanced product portfolio**
- **Beauty** is the top-performing category ($5.55M)
- **Sports** generates the lowest revenue ($5.41M), despite similar order volume
- Average ratings (~3.0) and discounts (~13.3%) are consistent across all categories
- Discounts alone do not significantly increase category performance

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💳 Payment & Regional Analysis

- **Wallet payments** lead with ~20.3% share and highest revenue ($6.68M)
- UPI and Cash on Delivery also contribute significantly (~20% each)
- No single payment method dominates → diversified payment ecosystem
- **Middle East** is the top region ($8.30M), closely followed by North America
- Regional revenue differences are minimal (~2.3%), indicating a balanced global presence

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💸 Discount Impact Analysis

- Orders with **no discount generate highest revenue per order ($749)**
- High discounts (21–30%) reduce revenue significantly (~$521 per order)
- 66.9% of orders fall within the 1–20% discount range
- Deep discounting negatively impacts profitability
- Shifting customers from high-discount to moderate-discount tiers can improve revenue

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⭐ Ratings vs Revenue

- Customer ratings have **minimal impact on revenue**
- Revenue per order varies only slightly (~$653–$661 across ratings)
- Majority of orders come from **2–3 star rated products (~50%)**

- 5-star products contribute ~23.6% of total orders

- Purchasing decisions are influenced more by pricing and availability than ratings

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📊 Business Recommendations

🚀 Revenue Growth

- Introduce targeted campaigns during low-performing months (February, November)

📦 Category Strategy

- Improve performance of low-revenue categories (Sports, Electronics) through promotions and product improvements

💸 Discount Optimization

- Reduce deep discounting (21–30%)

- Focus on moderate discounts (10–20%) to balance revenue and volume

💳 Payment Strategy

- Promote high-value payment methods (Credit Cards) using cashback and offers

🌍 Regional Expansion

- Focus on improving sales in Europe through localized strategies

⭐ Product Improvement

- Improve 2–3 star rated products to increase customer satisfaction and retention

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📌 Conclusion

The analysis reveals a stable and well-balanced sales structure with opportunities for improvement in discount optimization, category performance, and regional growth. Strategic adjustments in pricing, marketing, and product quality can significantly enhance overall business performance.

📎 Note

For full visual analysis and dashboard, refer to the Power BI dashboard and project README.